Sangyu Xu 徐桑榆

Computational Behaviour Scientist

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Computational behaviour scientist with 15+ years building measurement tools for internal states: automated tracking, effect-size estimation, and high-dimensional behavioral benchmarking. Currently building agentic LLM evaluation workflows for scientific reasoning. Also works with electrophysiology time-series — spike sorting, LFP analysis, and signal conditioning.

EXPERIENCE

Senior Research Fellow — Duke-NUS Medical School, Singapore *2026–present*

Computational ethomics pipelines for internal state inference and epitranscriptomics workflows for native RNA modification mapping. Developing agentic LLM evaluation systems for scientific reasoning — grounded analytical outputs feed adversarial multi-agent critique and structured manuscript assessment pipelines.

Senior Research Fellow / Research Fellow — IMCB, A*STAR Singapore *2017–2026*

Wrote and won a SGD 300,000 National Medical Research Council Young Individual Research Grant as sole investigator; ran it for four years (2021–2025) with one research assistant, whom I trained in genetics, assay development, automatic behavior tracking and pose estimation, machine learning, and native RNA sequencing.

Co-developed DABEST 2.0, an estimation statistics framework for visualising effect sizes and uncertainty; predecessor has 2,000+ citations. Built DESTRA, a ground-truth-referenced benchmarking approach for high-dimensional behavioural states, showing that perturbations can shift individual metrics without recapitulating the full target state — presented as an invited talk at APDNC3 / RIKEN 2024.

Developed automated behavioural classification pipelines for large-scale ethological data; contributed to novel optogenetic silencing tools (*Nature Communications* 2024) and long-Stokes-shift fluorescent markers for two-photon imaging (*microPublication Biology* 2026).

Postdoctoral Associate — Tonegawa Lab, MIT *2016–2017*

Designed the first olfactory-cued rodent intertemporal choice task, translating a paradigm from human preference evaluation to allow precise neural recording at the decision point, including *in vivo* optrode electrophysiology of dorsal raphe neurons. Built a psychometric model predicting choice probability as a function of delay; dorsal raphe serotonergic activation increased patience while silencing drove impulsivity (*Current Biology* 2017, co-corresponding). Also contributed to studies of noradrenergic attentional circuits (*PNAS* 2020) and basolateral amygdala valence encoding (*Nature Neuroscience* 2016).

EDUCATION

PhD Systems Neuroscience — MIT *2016*

Supervisor: Susumu Tonegawa. Dissertation: *Control of Intertemporal Choice by Dorsal Raphe Serotonergic Neurons.*

BS Biology (Minor: Music) — UC San Diego *2007*

Supervisor: Charles Zuker. Undergraduate dissertation: A Novel Trans-Synaptic Neuronal Tracing Strategy.

SELECTED PUBLICATIONS

Xu S, Zhang X, Stewart JC, Ho J, Choudhury D, Wang Z, Claridge-Chang A. Contextualized phenovectors reveal a *Drosophila* serotonergic circuit controlling satiety. (in revision, *Science Advances*). bioRxiv 2026.

Zhang X, **Xu S**, Ho J, Stewart JC, Claridge-Chang A. A mushroom-body output neuron that mediates octopamine-driven and hunger-motivated feeding in *Drosophila*. bioRxiv 2026.

Xu S, Zhang X, Cheung KY, Mai Y, Wu Y, Claridge-Chang A. Long-Stokes-Shift mScarlet3 as a structural marker for two-photon imaging. *microPublication Biology* 2026. doi:10.17912/micropub.biology.002142.

Lu Z*, Anns J*, Mai Y*, Zhang R, Lian K, Lee NM, Hashir S, Wang L, Li Y, Castillo-Gonzalez AR, Ho J, Choi H, **Xu S*†**, Claridge-Chang A†. Getting over ANOVA: estimation graphics for multi-group comparisons. *Nature Methods* 2026. doi:10.1038/s41592-026-03187-7. (Co-corresponding.)

Ott S, **Xu S**, Lee N, Hong I, Anns J, Suresh DD, Zhang Z, Zhang X, Harion R, Ye W, Chandramouli V, Jesuthasan S, Saheki Y, Claridge-Chang A. Kalium channelrhodopsins effectively inhibit neurons. *Nature Communications* 2024; 15:3480.

Bari A, **Xu S**, Pignatelli M, Takeuchi D, Feng J, Li Y, Tonegawa S. Differential attentional control mechanisms by two distinct noradrenergic pathways. *PNAS* 2020; 117(46):29080–29089.

Xu S†, Das G, Hueske E, Tonegawa S†. Dorsal raphe serotonergic neurons control intertemporal choice under trade-off. *Current Biology* 2017; 27(22):3440–3451. (Co-corresponding.)

Kim J, Pignatelli M, **Xu S**, Itohara S, Tonegawa S. Antagonistic negative and positive neurons of the basolateral amygdala. *Nature Neuroscience* 2016; 19(12):1636–1646.

GRANTS

NMRC Open Fund – Young Individual Research Grant (OF-YIRG), sole investigator *SGD 300,000, 2021–2025*

Neural Correlates of Social Influence on Feeding Behaviour (OFYIRG20nov-0051). Submitted Nov 2020; funded.

Applications not funded

A*STAR Career Development Award, 2020 call, as PI — Targeting Neuromodulator Receptors and Transporters for Controlling Protein Feeding Behavior in Mated Females in *Drosophila melanogaster* and *Aedes aegypti*. SGD 300,000 requested; letter of intent Sep 2020; shortlisted and interviewed.

A*STAR Career Development Award, 1st call, as PI — Investigating serotonin neurons as a link between metabolic dysfunction and dementia in *Drosophila melanogaster*. SGD 480,000 requested; letter of intent Jan 2020, rebuttal Mar 2020.

L'Oréal Singapore For Women in Science National Fellowship 2019, Life Sciences — applied Sep 2019.

RECOGNITION

Outstanding Short Talk Award — Neuroscience Singapore 2025

Young Individual Research Grant (SGD $300,000) — National Medical Research Council, Singapore, 2021–2025

Walle Nauta Award for Excellence in Graduate Teaching — MIT, 2011

National Science Scholarship — A*STAR, Singapore, 2004–2013

Honourable Mention — Singapore Physics Olympiad, 2002

TECHNICAL SKILLS

Python · R · Git · nbdev · Quarto · pandas · NumPy · SciPy · PyTorch · matplotlib · LLM-as-Judge · DSPy · LangGraph · Claude API · agentic evals · adversarial red-teaming · bootstrap CI · effect size estimation · uncertainty quantification · causal inference · A/B testing · experimental design · Oxford Nanopore · direct RNA-seq · epitranscriptomics · DeepLabCut · SLEAP · SQL

TEACHING

Modern Data Analysis for Biomedical Research (Teaching Assistant, recurring) *Duke-NUS, 2021–2023*

GMS6812 Modern Data Analysis for Biomedical Research — Feb 2022, Feb 2023

Evidence Session: Modern Data Analysis for Biomedical Research — Oct 2021, Oct 2022

Graduate and undergraduate courses (Teaching Assistant) *MIT, 2009–2011*

Walle Nauta Award for Excellence in Graduate Teaching, MIT, 2011

9.013 Molecular and Cellular Neuroscience II — Spring 2011

9.301 Neural Plasticity in Learning and Memory — Spring 2010

9.150 Biochemistry and Pharmacology of Synaptic Transmission — Fall 2009

YSC4229 Molecular Neuroscience — Genes, Brains, and Behaviour (Guest Lecturer) *Yale-NUS, Mar 2021, Mar 2023*

MENTORING

Hazimah Yusoff (Research Assistant, NMRC YIRG, 2022–2025) — genetics, assay development, automated behaviour tracking and pose estimation, machine learning, native RNA sequencing ⟶ National University Health System (NUHS)

John Lovett (summer attachment student, Jul–Aug 2012) — expression of protein in mouse brain

Sinem Kadıoğlu (postgraduate attachment student, Jun–Oct 2018) — development of novel neurogenetics tools ⟶ Scientist, Genome Institute of Singapore, A*STAR

Farhana Haque Anisha (undergraduate student, May–Aug 2018) — development of novel fly feeding assay ⟶ Analyst, Costello Medical

Zoe Ang (polytechnic intern, Jul 2018–Jan 2019) — development of novel fly feeding assay

Jade Oh (summer intern, Jun–Aug 2019) — development of novel neurogenetics tools ⟶ Clarendon Scholar, Oxford

Beatrix Tang (undergraduate intern, Jun–Nov 2021) — development of novel neurogenetics molecular tools ⟶ medical student, Duke-NUS Medical School, Singapore

SERVICE

Society for Neuroscience, Singapore Chapter *Singapore, 2018–2024*

Treasurer — 2021–2024

Director of Membership — 2018–2020

Journal reviewing — *Scientific Reports*

Outreach and public engagement

Drawing Neurons, art–science collaboration (scientist collaborator) — exhibited at the Neuroscience Singapore Symposium and The Substation, Singapore, 2019

Brain Festival and Brain Awareness Week (exhibitor) — live demonstration of *Drosophila* culture under light microscopes, Science Centre Singapore, 2018